Theorem 10.3

Let u and v be two differentiable functions. Then $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$.

Proof

Let u and v be the given two differentiable functions so that

$$\lim_{\Delta x\to 0}\frac{u(x+\Delta x)-u(x)}{\Delta x} = \frac{du}{dx} \text{ and } \lim_{\Delta x\to 0}\frac{v(x+\Delta x)-v(x)}{\Delta x} = \frac{dv}{dx}.$$

Let $y = f(x) = u.v$

Then $f(x + \Delta x) = u(x + \Delta x)\, v(x + \Delta x)$, and

$f(x + \Delta x) - f(x) = u(x + \Delta x)\, v(x + \Delta x) - u(x).v(x)$

$= v(x + \Delta x)\,[u(x+\Delta x)-u(x)] + u(x)[v(x+\Delta x)-v(x)]$.

This implies, $\frac{f(x+\Delta x)-f(x)}{\Delta x} = u(x)\frac{[v(x+\Delta x)-v(x)]}{\Delta x} + v(x+\Delta x)\frac{[u(x+\Delta x)-u(x)]}{\Delta x}$.

$$\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x} = u(x)\lim_{\Delta x\to 0}\frac{v(x+\Delta x)-v(x)}{\Delta x} + \lim_{\Delta x\to 0} v(x+\Delta x)\lim_{\Delta x\to 0}\frac{u(x+\Delta x)-u(x)}{\Delta x}$$

$= u(x)v'(x) + v(x)u'(x)$ (since v is continuous, $\lim_{\Delta x\to 0} v(x+\Delta x) = v(x)$.

That is, $f'(x) = \boxed{\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}}$.

or $\boxed{(uv)' = uv' + vu'}$

Similarly $(uvw)' = uvw' + uv'w + u'vw$.

This can be extended to a finite number of differentiable functions $u_1, u_2, \ldots u_n$, using induction :

$(u_1.u_2 \ldots, u_n)' = u_1u_2 \ldots u_{n-1}u_n' + u_1u_2 \ldots u_{n-1}'u_n + \cdots + u_1'u_2 \cdots u_n$.

Theorem 10.4 (Quotient Rule)

Let u and v be two differentiable functions with $v(x) \neq 0$. Then $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$.

Proof

Let $y = f(x) = \frac{u}{v}$, u and v are differentiable functions of x and where $v(x) \neq 0$.

Now $f(x+\Delta x) = \frac{u(x+\Delta x)}{v(x+\Delta x)}$

This implies, $f(x+\Delta x) - f(x) = \frac{u(x+\Delta x)}{v(x+\Delta x)} - \frac{u(x)}{v(x)}$

$= \frac{v(x)u(x+\Delta x) - u(x)v(x+\Delta x)}{v(x+\Delta x)v(x)}$.

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